

FIGURE CAPTIONS — Cross-Species CIS Study (submission edition, 2026-09-26)

Captions for the figure sequence defined in 10_REPORT/FIGURE_PROVENANCE.md. Every number is quoted from frozen artifacts; nothing was recomputed. File numbering vs paper numbering is documented in the provenance table.

Main figures

Figure 1. Study design and pipeline. From the population-scale AOMIC-ID1000 structural-connectome resource (900 subjects acquired; 801 QC-pass on frozen, CIS-blind flags), exact per-node Control Impact Scores were computed for all 456 nodes in every subject, followed by degree-matched controls ($\pm 10\%$ total degree), degree-preserving (Maslov–Sneppen) null ensembles in two batteries, BH-FDR node-level inference, two-level system-enrichment tests, a 9-configuration robustness ladder, and a pre-registered cross-scale comparison with the frozen fly (FAFB v783) study. (File: *fig01_workflow.png/pdf*.)

Figure 2. CIS architecture and its degree relation (801 subjects). (A) Population-mean CIS across all 456 nodes; CIS is unitless — the fraction of baseline global efficiency $E(G)$ lost on node removal — with a maximum population-mean CIS of 0.00462 (node 414). (B) CIS tracks node degree (median Spearman $\rho = 0.943$) far more closely than strength ($\rho = 0.486$), motivating the degree-controlled residual analysis of Fig. 4. (Files: *fig03_population_cis.png/pdf*, *fig04_cis_vs_degree_strength.png/pdf*.)

Figure 3. Degree-preserving null arbitration. (A) Per-node Stouffer z against the 100-null battery (battery A): 43 of 456 nodes carry finite z (maximum 23.30, node 400); the remaining 413 nodes sit at the add-one empirical- p floor and are explicitly counted as censored ($-\infty$), not hidden. (B) Per-subject top-50 concentration z (battery B): all 200/200 subjects lie above their null ensembles. All per-subject empirical p -values sit at the $1/101$ null-resolution floor; inference therefore rests on the z statistics and the population sign test (two-sided $p = 1.24 \times 10^{-60}$). (Files: *fig06a_stouffer.png/pdf*, *fig06b_battery_b.png/pdf*.)

Figure 4. Degree-independent residual architecture (801 subjects). Subject-median CIS residual against degree- and strength-matched controls (36,846 pairs), and the per-subject Cliff's δ distribution: the residual is positive in 778/801 subjects (sign $p = 2.6 \times 10^{-197}$) with median $\delta = 0.100$ [IQR 0.078–0.129]; 58.8% of individual matched comparisons beat the control, and the median paired Wilcoxon p is 0.167 — small in effect size, directionally near-universal across subjects. (File: *fig05_degree_controlled.png/pdf*.)

Figure 5. Anatomical distribution of the 43 FDR-surviving nodes. System counts of the 43/456 nodes surviving BH-FDR $q < .05$ (maximum Stouffer $z = 23.30$): Subcortical/Cerebellar 39, Limbic 2, Default 2. The pre-registered system-enrichment gate (R2b) itself was negative: visual cortex holds 1 of the top 50 (2%), and the single nominal system signal (Subcortical/Cerebellar, $z_B = 2.08$, $p = .031$, $1.25\times$) is degree-anchored and absent at $K = 25$. Rendered from the frozen *table_07_null_results.csv* plus the atlas label manifest; no statistics were recomputed. (File: *fig05_fdr_anatomical.png/pdf*.)

Figure 6. Cross-scale comparison (human vs fly), normalized comparators only. Effect-size (Cliff's δ), enrichment (z_B), and concentration comparators, normalized per the cross-scale discipline tags: human $\delta = 0.100$ vs fly $\delta = 0.0979$ (the fly value did not survive its own degree-preserving null ensemble, $p = 0.109$); human visual share of the top 50 is 2% versus 80% visual-centrifugal in the fly. Absolute CIS values are never compared across species (different units, graphs, and designs). (File: *fig08_cross_scale.png/pdf*.)

Supplementary figures

Figure S1. Population QC (900 subjects). QC flag distributions for the full acquired cohort; 99 subjects carry flags (94 isolated-node, 5 strength-IQR), retained rather than deleted; 801 subjects pass. Cohort selection was frozen before any CIS computation. (*File: fig02_population_qc.png/pdf.*)

Figure S2. Robustness ladder (9 configurations, n = 150 per configuration). Top-50 mean-of-means and Gini across atlases {4S256, 4S156, Brainnetome246Ext, AAL116}, weights {sift_invnodevol, radius2_count}, and costs {0.10, 0.20, 0.25}. All 9 configurations reproduce the primary architecture; the maximum like-for-like top-50-mean deviation is 0.00262 (atlas_AAL116; no numeric tolerance was pre-registered for E06 — descriptive comparison). (*File: fig07_robustness.png/pdf.*)

Figure S3. Conceptual schematic. Two-layer architecture: a dominant connectivity-dependent component plus a smaller degree-independent residual, arbitrated by matched controls and degree-preserving nulls. Schematic only; no data are plotted. (*File: fig09_conceptual.png/pdf.*)